

uc3m-thesis-ieee

Package Documentation — v0.4.0

Universidad Carlos III de Madrid thesis template, IEEE style

1. Configuration

The template exports a single `conf` function, which you apply to your document with `#show: conf.with(...)`.

1.1. Minimal example

codly

```
1  #import "@preview/uc3m-thesis-ieee:0.4.0": conf
2
3  #show: conf.with(
4    title: "My Bachelor Thesis",
5    author: "GUL UC3M",
6    degree: "Computer Science and Engineering",
7    advisors: ("Prof. L",),
8    location: "Leganés, Madrid",
9    thesis-type: "TFG",
10   date: datetime(year: 2025, month: 6, day: 15),
11   language: "en",
12   abstract: (
13     body: [A short description of my thesis.],
14     keywords: ("Keyword 1", "Keyword 2", "Keyword 3"),
15   ),
16   genai-declaration: (usage: false),
17 )
18
19 = Introduction
20
21 My thesis starts here.
```

1.2. Abstract

The `abstract` parameter takes a dictionary with the following keys:

Key	Type	Description
<code>body*</code>	<code>content</code>	The abstract text.
<code>keywords*</code>	<code>array of str</code>	Between 2 and 5 keywords. See IEEE Taxonomy.

```
1  abstract: (
2    body: [
3      This thesis presents an implementation of interrupts and timers
```

```

4     in the CREATOR simulator.
5 ],
6 keywords: ("Operating Systems", "Interrupts", "RISC-V", "Simulation"),
7 ),

```

For Spanish documents, it is required to provide an english-abstract with the same structure:

```

1 language: "es",
2 abstract: (
3     body: [Una breve descripción de mi tesis.],
4     keywords: ("Sistemas Operativos", "Interrupciones", "RISC-V"),
5 ),
6 english-abstract: (
7     body: [A brief description of my thesis.],
8     keywords: ("Operating Systems", "Interrupts", "RISC-V"),
9 ),

```

1.3. Epigraph

An optional opening quote placed before the abstract. Provide a dictionary with:

Key	Type	Description
quote*	content	The quote text.
author*	str	The author of the quote.
source	str	(Optional) The source where the quote was found.

```

1 epigraph: (
2     quote: [The purpose of abstracting is not to be vague, but to create a new
3         semantic level in which one can be absolutely precise.],
4     author: "Edsger W. Dijkstra",
5     source: "The Humble Programmer, 1972",
6 ),

```

1.4. Outlines

The outlines parameter accepts an optional dictionary with the following boolean keys. All keys are optional (default false):

Key	Type	Description
figures	bool	Include a list of figures.
tables	bool	Include a list of tables.
listings	bool	Include a list of code listings.
custom	array of content	Additional custom outlines (e.g. algorithms).

```

1 outlines: (
2     figures: true,
3     tables: true,
4     listings: false,
5     // custom: (

```

```

6 // outline(
7 //   title: [List of Algorithms],
8 //   target: figure.where(kind: "algorithm"),
9 // ),
10 // ),
11 ),

```

1.5. Abbreviations

You can pass abbreviations as a dictionary mapping abbreviations to their expansions (automatically sorted alphabetically):

```

1 abbreviations: (
2   API: "Application Programming Interface",
3   CPU: "Central Processing Unit",
4   ML: "Machine Learning",
5   TFG: "Trabajo de Fin de Grado",
6 ),

```

Or as custom content if you need full control over the layout:

```

1 abbreviations: [
2   #table(
3     columns: (auto, lfr),
4     [*API*], [Application Programming Interface],
5     [*ML*], [Machine Learning],
6   )
7 ],

```

1.6. Glossary

The template integrates with `glossarium` for automatic glossary and acronym management.

Define your entries in a separate file (e.g. `config/glossary.typ`):

```

1 #let glossary-entries = (
2   (
3     key: "API",
4     short: "API",
5     long: "Application Programming Interface",
6     description: "A set of protocols and tools for building software.",
7   ),
8   (
9     key: "ML",
10    short: "ML",
11    long: "Machine Learning",
12    description: "A subset of AI that enables computers to learn from
13      data.",
14  ),
15 )

```

Then import it in your main file and pass it to conf:

```
1 #import "config/glossary.typ": glossary-entries
2
3 #show: conf.with(
4   // ...
5   glossary: glossary-entries,
6 )
```

To reference a glossary term in the body text, use `#gls("key")` (singular) or `#glspl("key")` (plural). These are re-exported from `glossarium` and available after the `#show: conf.with(...)` call:

```
1 #import "@preview/uc3m-thesis-ieee:0.4.0": conf
2 #import "@preview/glossarium:0.5.9": gls, glspl
3
4 // In the body:
5 The #gls("API") exposes several endpoints.
6 Multiple #glspl("ML") models were evaluated.
```

Alternatively, you can pass fully custom content to `glossary` if you don't want to use `glossarium`.

1.7. Generative AI Declaration

The `genai-declaration` parameter is **required** by the university. You can either:

1.7.1. Option A: Use the built-in template (recommended)

If you did not use generative AI (`usage: false`):

```
1 genai-declaration: (usage: false),
```

If you did use generative AI (`usage: true`), it is required to provide the full declaration:

Pass a dictionary with the following keys:

Key	Type	Required	Description
<code>usage*</code>	<code>bool</code>	Always	Whether you used generative AI in the thesis.
<code>data-usage</code>	<code>dictionary</code>	If <code>usage</code> is true	Data usage declaration. See below.
<code>technical-usage</code>	<code>dictionary</code>	If <code>usage</code> is true	Technical usage declaration. See below.
<code>usage-reflection</code>	<code>content</code>	If <code>usage</code> is true	Personal reflection on AI usage.

data-usage keys — all are `bool` (`true` = YES, `false` = NO):

Key	Description
<code>confidential</code>	Did you submit confidential data (trade secrets, proprietary info)?
<code>copyright</code>	Did you submit copyrighted materials?
<code>personal</code>	Did you submit personal data?
<code>tos</code>	Did your usage respect the tool's terms of use?

technical-usage keys — include only those that apply, with a brief description of the prompt and interaction. Items are automatically grouped into two sections matching the official form:

Documentation and drafting: reflection, review, research, references, summary, translation

Develop specific content: assistance-coding, generating-content, optimization, data-processing, idea-inspiration, other

An optional tool key (str) declares the AI system name and version (e.g. "ChatGPT 4o").

```

1  genai-declaration: ( typ
2      usage: true,
3      data-usage: (
4          // true = YES / used with authorization; false = NO / not used
5          confidential: false,
6          copyright: false,
7          personal: false,
8          tos: true,
9      ),
10     technical-usage: (
11         // Name of the AI system and version (optional):
12         // tool: "ChatGPT 4o",
13
14         // — Documentation and drafting —————
15         reflection: [I asked for a list of alternatives to address the
16             problem stated at the beginning of the paper.],
17         // review: [...],
18         // research: [...],
19         // references: [...],
20         // summary: [...],
21         // translation: [...],
22
23         // — Develop specific content —————
24         assistance-coding: [I asked for explanations about a Python code
25             snippet and for help correcting an error.],
26         // generating-content: [...],
27         // optimization: [...],
28         // data-processing: [...],
29         // idea-inspiration: [...],
30         // other: [...],
31     ),
32     usage-reflection: [
33         Overall, I found generative AI useful for boilerplate tasks and
34         quick explanations, but it required careful verification of all
35         generated content. It did not replace critical thinking or
36         domain-specific knowledge.
37     ],

```

```
38 ),
```

1.7.2. Option B: Provide custom content

If you need full control, pass any content directly:

```
1 genai-declaration: [ typ
2   // Your custom AI declaration content here.
3 ],
```

1.8. Bibliography

Pass the result of Typst's built-in `bibliography()` call:

```
1 // config/bibliography.typ typ
2 #let bibliography-file = "../references.bib"
3 #let bibliography-style = "ieee"
4 #let bibliography = bibliography(bibliography-file, style: bibliography-
  style)
```

```
1 // report.typ typ
2 #import "config/bibliography.typ": bibliography
3
4 #show: conf.with(
5   // ...
6   bibliography-content: bibliography,
7 )
```

2. Styles

The template offers three visual styles, controlled by the `style` parameter.

2.1. "fancy" (default)

An opinionated style inspired by the original LaTeX template, with:

- Decorative chapter title pages with a horizontal rule and chapter number/name centered
- Serif headers showing the current chapter name and page number
- IEEE-style figure captions and table captions
- **Libertinus Serif** body font

uc3m

Universidad Carlos III de Madrid

Grado en Ingeniería Tipstática

Trabajo de Fin de Grado

«Análisis, diseño, e implementación del mejor Trabajo de Fin de Grado de la historia»

Autor

Nombre Extremadamente Largo e Incómodo de Escribir

Tutor

Profesor Cuyos Padres Tenían Visión de Futuro

Leganés, Madrid

Abril 2025



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Figure 1: Example of the **fancy** style.

2.2. "clean"

A clean, minimal style inspired by clean-dhbw, with:

- Large chapter number in the background on chapter pages
- Simple running header showing the chapter name
- Clean-style figure and table captions
- **Libertinus Serif** body font

uc3m

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A. Declaración de uso de IA generativa

Figure 2: Example of the **clean** style.

2.3. "strict"

Strictly follows the UC3M library guidelines:

- Times New Roman font, 12pt
- Centered uppercase chapter headings
- Page numbers in the footer (Roman numerals in front matter, Arabic in body)
- No decorative elements
- Double-sided layout is **not** permitted with this style



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Figure 3: Example of the **strict** style.

> [!NOTE] > You can mix styles for the title page and body independently using `titlepage-style`. For example, to use a fancy title page with a strict body: `style: "strict", titlepage-style: "fancy"`.

2.4. All parameters (API Reference)

lib

- `conf()`
- `outline-entry-formatter()`

conf

Main configuration function.

Recommended to use with `#show`: `conf.with(...)`.

Parameters

```

conf(
  doc: content,
  title: str,
  author: str,
  degree: str,
  advisors: array,
  location: str,
  thesis-type: str,
  date: datetime,
  bibliography-content: content,
  language: str,
  style: str,
  titlepage-style: str auto,
  table-style: str auto,
  figure-style: str auto,
  figure-spacing: length none,
  double-sided: bool,
  logo: str,
  short-title: str,
  date-format: str auto,
  license: bool,
  flyleaf: bool,
  epigraph: dictionary none,
  abstract: dictionary,
  english-abstract: dictionary,
  acknowledgements: content none,
  outlines: dictionary none,
  appendixes: content none,
  glossary: array content none,
  abbreviations: dictionary content none,
  genai-declaration: dictionary content
) -> content

```

doc content

Thesis contents.

title str

Title of the thesis.

Default: none

author str

Author name.

Default: none

degree str

Degree name (e.g. "Computer Science and Engineering").

Default: none

advisors array

List of advisor names.

Default: none

location str

Presentation location.

Default: none

thesis-type str

Type of thesis ("TFG" or "TFM").

Default: none

date datetime

Presentation date.

Default: none

bibliography-content content

Bibliography contents, usually calling bibliography.

Default: none

language str

"en" or "es".

Default: none

style `str`

Visual style, mainly affecting headings, headers, and footers. The available styles are `strict`, which strictly follow's the university library's guidelines, `clean`, based on `clean-dhbw`, and `fancy`, based on my original LaTeX version.

Default: `"fancy"`

titlepage-style `str` or `auto`

Style for the titlepage (see `style`). If set to `auto`, uses the main style.

Default: `auto`

table-style `str` or `auto`

Style for the table caption, either `"clean"` or `"ieee"`. If set to `auto`, uses the default for the main style (`"clean"` for `clean` style, `"ieee"` for the rest).

Default: `auto`

figure-style `str` or `auto`

Style for the figure caption, either `"clean"` or `"ieee"`. If set to `auto`, uses the default for the main style (`"clean"` for `clean` style, `"ieee"` for the rest).

Default: `auto`

figure-spacing `length` or `none`

Extra spacing to give to figures and tables. If `none`, no extra spacing.

Default: `0.75em`

double-sided `bool`

Whether to use double-sided pages. This is not allowed in the `strict` style.

Default: `false`

logo `str`

Type of logo (`"old"` or `"new"`).

Default: `"new"`

short-title `str`

Shorter version of the title, to be displayed in the headers. Only applies if `double-sided` is set to `true`.

Default: `none`

date-format `str` or `auto`

Date format. Use `auto` or specify the format using the [Typst format syntax](https://typst.app/docs/reference/foundations/datetime/#format).

Default: `auto`

license `bool`

Whether to include the CC BY-NC-ND 4.0 license.

Default: `true`

flyleaf `bool`

Whether to include a blank page after the cover.

Default: `true`

epigraph `dictionary` or `none`

A short quote that guided you through the writting of the thesis, your degree, or your life. Consists of quote (of type `content`), the body or text itself, author (of type `str`), the author of the quote and, optionally, source (of type `str`), where the quote was found.

Default: `none`

abstract `dictionary`

A short and precise representation of the thesis content. Consists of `body` (of type `content`), the main text, and keywords, an array of key terms (of type `str`) (see [IEEE Taxonomy](https://www.ieee.org/content/dam/ieee-org/ieee/web/org/pubs/ieee-taxonomy.pdf)).

Default: `none`

english-abstract `dictionary`

An english translation of the abstract. Compulsory for spanish works, invalid for english ones.

Default: `none`

acknowledgements `content` or `none`

Text where you give thanks to everyone that helped you.

Default: `none`

outlines `dictionary` or `none`

Set of extra outlines to include (figures, tables, listings), and extra custom outlines (custom, an array of contents – typically the result of calling `outline`).

Default: `none`

appendixes `content` or `none`

Set of appendixes.

Default: `none`

glossary `array` or `content` or `none`

Glossary entries. If `array` is provided, it will use the `glossarium` library. If `content` is passed, it will display that content, without applying any styling.

Default: `none`

abbreviations `dictionary` or `content` or `none`

Abbreviations, acronyms and initials used throughout the thesis. You can provide a map (dictionary of strings) or a custom one (`content`).

Default: `none`

genai-declaration `dictionary` or `content`

Information about the use of Generative AI in the thesis. You can supply your own content, or use the university's template, by supplying a dictionary. See the example for more details.

Default: `none`

outline-entry-formatter

Formats an outline entry.

Parameters

```
outline-entry-formatter(
  it: content,
  prefix: auto content,
  body: auto content,
  fill: boolean,
  page: boolean,
  above: auto fraction relative,
  spacing: auto fraction relative,
  text-size: auto length,
  text-weight: int str,
  color: color,
  indented: boolean,
  justified: boolean
) -> content
```

it `content`

Outline entry.

prefix `auto` or `content`

Entry prefix (e.g. [Chapter 1]). If auto, default prefix.

Default: `auto`

body `auto` or `content`

Entry body (e.g. [My chapter]). If auto, default body.

Default: `auto`

fill `boolean`

When true, add dots between the body and the page number.

Default: `true`

page `boolean`

When true, shows the page number.

Default: `true`

above `auto` or `fraction` or `relative`

The spacing between this block and its predecessor (`block.above`).

Default: `auto`

spacing `auto` or `fraction` or `relative`

The spacing around the block (`block.spacing`). When `auto`, inherits the paragraph spacing.

Default: `auto`

text-size `auto` or `length`

Text size. When `auto`, default text size. (`text.size`)

Default: `auto`

text-weight `int` or `str`

Text weight (`text.weight`).

Default: `"regular"`

color `color`

Entry text color. Includes link color.

Default: `black`

indented `boolean`

Whether to indent the entry.

Default: `true`

justified `boolean`

When `false`, wraps the entry in a block to prevent justification.

Default: `true`

generative-ai

- `genai-template()`

genai-template

UC3M template for declaration of use of AI in the project.

Parameters

```
genai-template(
    language: str,
    usage: bool,
    data-usage: dictionary,
    technical-usage: dictionary,
    usage-reflection: content
)
```

language `str`

The language of the project (e.g. “en”, “es”).

usage `bool`

Whether AI is used in the project.

data-usage `dictionary`

AI data usage declaration.

technical-usage `dictionary`

Technical usage of AI.

usage-reflection `content`

Reflection on the usage of AI.